

B11902004

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Q1

Binary search is $O(n)$ in the worst case because if the element is missing you may have to look at every cell to be sure. The recurrence just shows the best case is faster.

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Q2

I ran out of time — please see my Q2 answer which also covers this.
(arrow pointing off the page)

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Q3

(pencil smudge) ... midterm of the other course ... (crossed out)
see attached sheet (no sheet attached)

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Q4

True by induction.